JAVA PROGRAM REPORT

Digital Payment System

Programming Language:	Java
Topic:	User Input and Conditional Statements
Program File:	DigitalPayment.java

1. AIM

To create a simple digital payment program in Java that accepts a customer name, account balance, and payment amount, then checks whether the payment can be completed.

2. DESCRIPTION

The program uses Scanner to read input from the keyboard. If the payment amount is less than or equal to the account balance, the amount is deducted and payment details are displayed. Otherwise, the program displays a payment failure message due to insufficient balance.

3. ALGORITHM

1. Start the program and create a Scanner object.
2. Read the customer name.
3. Read the account balance and payment amount.
4. If amount ≤ balance, subtract the amount from the balance and display payment details.
5. Otherwise, display Payment Failed and Insufficient Balance.
6. Close the Scanner and stop.

4. SOURCE CODE

Save the code as **DigitalPayment.java**, matching the public class name.

```
import java.util.Scanner;

public class DigitalPayment {
    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.print("Enter Customer Name: ");
        String name = sc.nextLine();

        System.out.print("Enter Account Balance: ");
        double balance = sc.nextDouble();

        System.out.print("Enter Payment Amount: ");
        double amount = sc.nextDouble();

        if (amount <= balance) {
            balance = balance - amount;

            System.out.println("\n--- Payment Details ---");
            System.out.println("Customer Name: " + name);
            System.out.println("Payment Successful");
            System.out.println("Paid Amount: " + amount);
            System.out.println("Remaining Balance: " + balance);
        } else {
            System.out.println("Payment Failed");
            System.out.println("Insufficient Balance");
        }

        sc.close();
    }
}
```

5. PROGRAM EXPLANATION

1. Scanner: Reads customer details and numeric values entered by the user.

2. nextLine(): Reads the customer name, including spaces.

3. nextDouble(): Reads the account balance and payment amount as decimal numbers.

4. if (amount <= balance): Checks whether the account has enough money for the payment.

5. Balance update: If payment is possible, the amount is subtracted from the balance.

6. else block: Displays an error message when the payment amount exceeds the balance.

7. sc.close(): Closes the Scanner after input is complete.

6. EXECUTION AND OUTPUT

Compile and run:

```
C:\Java> javac DigitalPayment.java
C:\Java> java DigitalPayment
```

Test Case 1: Successful payment (sample input: Ravi, 5000, 1200)

```
Enter Customer Name: Ravi
Enter Account Balance: 5000
Enter Payment Amount: 1200

--- Payment Details ---
Customer Name: Ravi
Payment Successful
Paid Amount: 1200.0
Remaining Balance: 3800.0
```

6. EXECUTION AND OUTPUT (CONTINUED)

Test Case 2: Insufficient balance (sample input: Suresh, 1000, 1500)

```
Enter Customer Name: Suresh
Enter Account Balance: 1000
Enter Payment Amount: 1500
Payment Failed
Insufficient Balance
```

7. TEST CASE SUMMARY

Test Case	Input	Expected / Actual Result	Status
1. Successful payment	Balance 5000, amount 1200	Payment Successful; remaining balance 3800.0	PASS
2. Insufficient balance	Balance 1000, amount 1500	Payment Failed; Insufficient Balance	PASS
3. Exact balance	Balance 1000, amount 1000	Payment succeeds; remaining balance 0.0	Logic

8. RESULT

The program compiled and ran successfully for both successful and insufficient-balance cases. The conditional statement correctly determines whether the entered payment amount can be deducted from the account balance.

9. CONCLUSION

This program demonstrates keyboard input using Scanner, arithmetic operations, and if-else decision making in Java to simulate a basic digital payment transaction.